DESIGN AND FABRICATION OF A WASTE HEAT RECOVERY SYSTEM FROM ENGINE EXHAUST USING THERMOELECTRIC GENERATOR

PROJECTREPORT

Submitted by

KAMARAJ M	**8115U23ME021**
PRADEEP K	**8115U23ME034**
SANTHOSH M	**8115U23ME039**
SIVA K	**8115U23ME045**

***inpartialfulfilmentfortheawardofthecourse** of*

GEA1401–ENVIRONMENTALSCIENCE(TCPR)

IN DEPARTMENT OF
MECHANICALENGINEERING

K.RAMAKRISHNANCOLLEGEOFENGINEERING
(AUTONOMOUS)
SAMAYAPURAM,TRICH

ANNAUNIVERSITY
CHENNAI600025

NOV2026

DESIGN AND FABRICATION OF A WASTE HEAT RECOVERY SYSTEM FROM ENGINE EXHAUST USING THERMOELECTRIC GENERATOR

APROJECT REPORT

Submitted by

KAMARAJ M	**8115U23ME021**
PRADEEP K	**8115U23ME034**
SANTHOSH M	**8115U23ME039**
SIVA K	**8115U23ME045**

in partial fulfilment for the award of the course

of

GEA1401 – ENVIRONMENTAL SCIENCE (TCPR)

IN DEPARTMENT OF

MECHANICAL ENGINEERING

Under the Guidance of

Dr.S.NANDHA GOPAN M.E., Ph.D

K.RAMAKRISHNAN COLLEGEOF ENGINEERING
(AUTONOMOUS)
Under
ANNA UNIVERSITY, CHENNAI

K. RAMAKRISHNAN COLLEGEOFENGINEERING
(AUTONOMOUS)
Under
ANNAUNIVERSITY,CHENNAI

BONAFIDECERTIFICATE

Certified that this project report titled **"DESIGN AND FABRICATION OF A WASTE HEAT RECOVER SYSTE FROM EXHAUS ENGIN USIN THREMOELECTRI GENERATOR**" is the Bonafide work of **KAMARAJ M (8115U23ME021), PRADEEP K(8115U23ME034), SANTHOSH M (8115U23ME039), SIVA K (8115U23ME045),** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr.H. RAMAKRISHNAN,M.E.,Ph.D.,
HEADOFTHE DEPARTMENT,

ASSOCIATE PROFESSOR
DepartmentofMechanicalEngineering,
K.RamakrishnanCollegeEngineering,
Samayapuram, Trichy-621 112.

SIGNATURE

Dr.S.NANDHA GOPAN M.E., Ph.D.,

ASSISTANT PROFESSOR,
Departmentof MechanicalEngineering,
K.RamakrishnanCollegeEngineering,
Samayapuram, Trichy-621 112.

SIGNATURE OF INTERNAL EXAMINER
NAME:
DATE:

SIGNATUREOFEXTERNALEXAMINER
NAME:
DATE:

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)
Under
ANNA UNIVERSITY

CHENNAI DECLARATION BY THE CANDIDATES

We jointly declare that the project report on **"DESIGN AND FABRICATION OF A WASTE HEAT RECOVERY SYSTEM FROM EXHAUST ENGINE USING THREMOELECTRIC GENERATOR"** is the result of original work done by us and best of our knowledge, similar work has not been submitted to "ANNA UNIVERSITY CHENNAI" for the requirement of Degree of BACHELOR OF ENGINEERING. This project report is submitted on the partial fulfillment of the requirement of the award of degree of BACHELOR OF ENGINEERING.

SIGNATURE OF THE CANDIDATES

KAMARAJ M

PRADEEP K

SANTHOSH M

SIVA K

Place: Samayapuram

Date:

ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution, "**K. RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS)**", for providing us
with the opportunity to do this project.

I extend our sincere acknowledgment and appreciation to the esteemed and honorable Chairman, **Dr. K. RAMAKRISHNAN, B.E.,** for having provided the facilities during the course of our study in college.

I would like to express our sincere thanks to our beloved Executive Director, **Dr. S. KUPPUSAMY, MBA, Ph.D.,** for forwarding our project and offering an adequate duration to complete it.

I would like to thank **Dr. D. SRINIVASAN, B.E, M.E., Ph.D.,** Principal, who gave the opportunity to frame the project to full satisfaction.

I thank **Dr. H. RAMAKRISHNAN, M.E., Ph.D.,** Head of the Department, Mechanical Engineering, for providing his encouragement in pursuing this project. We wish to convey our profound and heartfelt gratitude to our esteemed project guide, **Dr.S.NANDHA GOPAN M.E., Ph.D** Professor Department of Mechanical Engineering, for his incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

I render our sincere thanks to the Course Coordinator **Dr.M.CHRISPIN DAS M.E.,Ph.D** Professor and other staff members for providing valuable information during the course.

I wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period of the work progress.

DEPARTMENT OF MECHANICAL ENGINEERING
Program: Bachelor of Engineering

VISION & MISSION OF THE COLLEGE

VISION:

- To achieve a prominent position among the top technical institutions

MISSION:

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To provide education that can be used as a tool to transform the students.

VISION & MISSION OF THE DEPARTMENT

VISION:

- To provide eminent Mechanical Engineers to the society and industries through quality technical education and research.

MISSION:

- To impart quality professional education through state-of-the-art-infrastructure and resources.
- To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.
- To carryout collaborative projects with industries and academics with fullest potential.

DEPARTMENT OF MECHANICAL ENGINEERING

Program: Bachelor of Engineering

Program Educational Objectives :(PEOs)

Graduates of Mechanical Engineering shall:

1. Have Successful careers in industries or opt for higher studies and research.

2. Analyze, synthesize and design mechanical and allied products addressing societal issues.

3. Exhibit technical expertise, communication skills, team work with professionalism with ethical values.

Program Specific Outcomes (PSOs)

Students shall:

PSO1: Identify, formulate and solve engineering problems in three core streams of Mechanical Engineering, i.e., Design, Thermal and Manufacturing Engineering

PSO2: Design, develop, test and implement energy efficient systems for required Engineering applications.

PROGRAM OUTCOMES

PROGRAM OUTCOMES (PO's)	
PO 1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledg and research methods including design of experiments, analysis an interpretation of data, and synthesis of the information to provide vali conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PROGRAM's OUTCOME MAPPING

Check	PO number	Explanation
☑	**PO1 – Engineering Knowledge**	Applied heat transfer, airflow, and mechanical design principles to develop the grain dryer.
☑	**PO2 – Problem Analysis**	Identified grain-drying challenges and considered ways to reduce moisture and post-harvest losses.
☑	**PO3 – Design/Development of Solutions**	Designed a dryer to meet grain-processing needs while considering safety and usability.
☑	**PO4 – Conduct Investigations**	Tested the dryer and evaluated temperature, drying time, and grain condition.
☑	**PO5 – Modern Tool Usage**	Used suitable design, fabrication, and monitoring tools during project development.
☑	**PO6 – Engineer and Society**	Considered farmers' needs and the safe use of grain-drying equipment.
☑	**PO7 – Environment and Sustainability**	Aimed to reduce grain wastage and support sustainable use of harvested produce.
☑	**PO8 – Ethics**	Followed responsible engineering practices and reported project results honestly.
☑	**PO9 – Individual and Team Work**	Contributed to the team's design, fabrication, and project activities.
☑	**PO10 – Communication**	Presented the design, working method, and results through reports and presentations.
☑	**PO11 – Project Management and Finance**	Planned project tasks, selected components, and considered the project budget.
☑	**PO12 – Life-long Learning**	Learned about grain-drying technology and considered opportunities for future improvement.

SUSTAINABLE DEVELOPMENT GOAL



PRIMARY SDG's USED IN THE PROJECT



SDG 7 – AFFORDABLE AND CLEAN ENERGY

SDG 7 focuses on generating electrical energy from waste heat released by engine exhaust systems. It uses a thermoelectric generator to convert thermal energy into electrical energy through the Seebeck effect. This approach explores an alternative method of energy recovery without requiring additional fuel for the heat source.

SECONDARY SDG's USED IN THE PROJECT



SDG 9 – INDUSTRY, INNOVATION AND INFRASTRUCTURE

SDG 9 introduces an innovative method of recovering waste heat using thermoelectric technology. It combines a thermoelectric module, aluminium heat sink, heat collection arrangement, and electrical monitoring components. The prototype demonstrates how engineering technology can be developed to utilize otherwise wasted thermal energy.



SDG 12 – RESPONSIBLE CONSUMPTION AND PRODUCTION

SDG 12 focuses on utilizing waste thermal energy that would otherwise be released into the surrounding environment. By investigating the conversion of waste heat into electrical energy, it promotes better utilization of available energy resources. The thermoelectric system demonstrates a potential approach to reducing energy wastage in engines and industrial equipment.

ABSTRACT

The increasing demand for energy and the need for improved fuel efficiency have encouraged the development of alternative energy recovery technologies. Internal combustion engines release a considerable amount of thermal energy through exhaust gases, which is normally discharged into the atmosphere without being utilized. The project titled “Design and Fabrication of a Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator” focuses on investigating the possibility of converting waste thermal energy into useful electrical energy using thermoelectric technology.

The proposed system operates on the Seebeck effect, in which an electrical voltage is generated when a temperature difference is maintained between the hot and cold surfaces of a thermoelectric module. The experimental prototype consists of a TEC1-12706 thermoelectric module, a heat collection plate, and electrical measurement components. Thermal energy is transferred to the hot side of the module through the heat collection arrangement, while the aluminium heat sink dissipates heat from the cold side. The resulting temperature difference enables the generation of direct-current electrical voltage without moving mechanical parts.

During preliminary laboratory testing, a candle flame was used as the heat source to demonstrate the basic working principle of the system. An output voltage of approximately 0.2 V to 0.3 V was observed, demonstrating the conversion of a temperature difference into electrical voltage. The performance of the system depends on factors such as heat transfer efficiency, temperature difference, thermal contact, and cooling effectiveness.

The project demonstrates the feasibility of thermoelectric voltage generation at the laboratory level. Although the present prototype has not been tested directly on an engine exhaust system, it provides a foundation for future improvements using high-temperature thermoelectric generator modules, efficient heat exchangers, and improved cooling arrangements. The proposed concept has potential applications in automotive and industrial waste heat recovery, contributing to better energy utilization and sustainable engineering practices.

Keywords: *Waste Heat Recovery, Thermoelectric Generator, Engine Exhaust, Seebeck Effect, TEC1-12706, Aluminium Heat Sink, Thermal Energy, Electrical Power Generation*.

TABLE OF CONTENT

CHAPTER NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	**v**
	SUSTAINABLE DEVELOPMENT GOALS	**xi**
	ABSTRACT	**xii**
	TABLE OF CONTENT	**xiii**
	LIST OF FIGURES	**xvii**
	LIST OF ABBREVATIONS	**xviii**
	LIST OF TABLES	**xix**
1	**INTRODUCTION**	**1**
	1.1 GENERAL INTRODUCTION	1
	1.2 FEATURES	2
2	**LITERATURE REVIEW**	**3**
	2.1 WASTE HEAT RECOVERY FROM INTERNAL COMBUSTION ENGINES	3
	2.2 THERMOELECTRIC GENERATOR MATERIALS AND PERFORMANCE	3
	2.3 HEAT TRANSFER ENHANCEMENT IN THERMOELECTRIC GENERATORS	4
	2.4 THERMAL MANAGEMENT AND COOLING ARRANGEMENTS	4
	2.5 AUTOMOTIVE THERMOELECTRIC GENERATOR OPTIMIZATION	5

3	**PROBLEM IDENTIFICATION**	**6**
	3.1 LOSS OF THERMAL ENERGY THROUGH EXHAUST GASES	6
	3.2 LIMITED CONVERSION OF WASTE HEAT IN TO USEFUL ENERGY	6
	3.3DIFFICULTY IN MAINTAINING A TEMPERATURE DIFFERENCE	6
	3.4 LIMITATIONS OF THERMOELECTRIC MODULE PERFORMANCE	7
	3.5 CHALLENGES IN ELECTRICAL OUTPUT MEASUREMENT	7
	3.6 THERMAL PROTECTION AND COOLING REQUIREMENTS	7
4	**OBJECTIVE**	**8**
	4.1 GENERAL OBJECTIVE	8
	4.2 SPECFIC OBJECTIVE	8
5	**SELECTION OF MATERIALS**	**10**
	5.1 THERMOELECTRIC MODULE(TEC1-12706)	10
	5.2 ARDUINO MICROCONTROLLER	10
	5.3 TEMPERATURE SENSORS	11
	5.4 VOLTAGE MEASUREMENT CIRCUIT	11
	5.5 LCD DISPLAY	12
	5.6 DC–DC CONVERTER	**13**
	5.7 ELECTRICAL CONNECTIONS AND WIRING	13
	5.8 SUPPORTING FRAME AND ASSEMBLY	14
	5.9 POWER SUPPLY AND PROTECTION COMPONENTS	**14**

6	**COMPONENTS**	**15**
	6.1 **COMPONENTS**	**15**
7	**METHODOLOGY**	**17**
	7.1 WORKING METHOD	17
	7.2 WORKING METHOD FLOWCHART	18
8	**COST ESTIMATION**	**19**
	8.1 BILL OF MATERIALS	19
	8.2 TOTAL COSTESTIMATION	19
9	**WORKING PROCESS**	**20**
	9.1 OVERVIEW OF THE WORKING PROCESS	20
	9.2 STEP-BY-STEP OPERATION	20
	9.3 ELECTRICAL OUTPUT MEASUREMENT	22
	9.4 APPLICATIONS AND LIMITATIONS	23
	9.5 CIRCUIT DIAGRAM	24
	9.6 STEP BY STEP PROCESS	24
10	**FABRICATED MODEL**	**25**
	10.1 FABARICATED MODEL	25
11	**DESIGN CALCULATION**	**26**
	11.1 TEMPERATURE DIFFERENCE CALCULATION	26
	11.2 SEEBECK VOLTAGE CALCULATION	26
	11.3 ELECTRICAL POWER CALCULATION	26
	11.4 MAXIMUM POWER OUTPUT CALCULATION	27
	11.5 THERMOELECTRIC EFFICIENCY CALCULATION	27

	11.6 CALCULATION RESULT	28
12	**CONCLUSION**	**29**
	12.1CONCLUSION OF THE PROJECT	29
13	**FUTURE SCOPE**	31
	13.1 FUTURE SCOPE OF THE PROJECT	31
14	**REFERENCES**	**33**
	14.1 BOOKS	33
	14.2 RESEARCH PAPERS/JOURNALS	33

LIST OF FIGURES

S.NO	FIGURE NO	TITLE	PAGE NO
1	6.1	THERMOELECTRIC GENERATOR(TEG)	15
2	6.2	ARDUINO UNO BOARD	15
3	6.3	DC– DC CONVERTER	16
4	6.4	LCD DISPLAY	16
5	6.5	VOLTAGE SENSOR	16
6	7.1	WORKING METHOD FLOW CHART	18
7	9.1	CIRCUIT DIGRAM	23
8	9.2	STEP BY STEP PROCESS	23
9	10.1	FABRICATED MODEL	24

LIST OF ABBREVATION

S. No.	Abbreviation	Full Form
1	TEG	Thermoelectric Generator
2	TEC	Thermoelectric Cooler
3	DC	Direct Current
4	AC	Alternating Current
5	DC–DC	Direct Current to Direct Current
6	EMF	Electromotive Force
7	MCU	Microcontroller Unit
8	LCD	Liquid Crystal Display
9	USB	Universal Serial Bus
10	PCB	Printed Circuit Board
11	PWM	Pulse Width Modulation
12	ADC	Analog-to-Digital Converter

LIST OF TABLES

S.NO	TABLE NO	TITLE	PAGE NO
1	8.1	COST ESTIMATION	19
2	11.2	CALCULATION RESULT	28

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The increasing demand for energy-efficient automotive systems has created a need for technologies that can reduce energy losses and improve overall engine performance. Internal combustion engines are widely used in automobiles, transportation, and industrial applications. However, these engines convert only a portion of the chemical energy supplied by fuel into useful mechanical work. A significant amount of energy is lost in the form of heat through exhaust gases, cooling systems, and other engine components. This wasted thermal energy reduces the overall efficiency of the engine and contributes to environmental heat pollution.

Waste heat recovery is an effective approach for utilizing thermal energy that would otherwise be released into the surroundings. One of the promising technologies for recovering waste heat is the Thermoelectric Generator (TEG), which converts thermal energy directly into electrical energy. The working principle of a thermoelectric generator is based on the Seebeck effect. When a temperature difference is maintained between the hot and cold sides of a thermoelectric module, an electrical voltage is generated. This process enables electricity generation without moving mechanical components.

The proposed project, “Design and Fabrication of a Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator,” focuses on developing a prototype to demonstrate the conversion of waste thermal energy into useful electrical energy. The system utilizes a TEC1-12706 thermoelectric module positioned between a heat-collecting surface and an aluminium heat sink.

An Arduino microcontroller is incorporated into the proposed system for monitoring electrical parameters. The generated electrical voltage can be measured and displayed using suitable sensing and display components. A DC–DC converter can also be used to regulate the generated voltage for low-power auxiliary applications, depending on the available electrical output. The system is designed with a compact structure, simple mechanical arrangement, and minimal maintenance requirements.

1.2 FEATURES

The proposed Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator incorporates several important features that support thermal energy utilization, electrical power generation, and experimental monitoring.

1.WASTE HEAT RECOVERY

The system is designed to utilize thermal energy that would otherwise be released through engine exhaust gases. Recovering a portion of this wasted heat provides an opportunity to improve overall energy utilization without directly consuming additional fuel for electrical generation.

2. THERMOELECTRIC POWER GENERATION

The system uses a TEC1-12706 thermoelectric module to demonstrate electrical power generation through the Seebeck effect. A temperature difference between the hot and cold sides of the module produces a direct-current voltage without requiring moving mechanical components.

3. ARDUINO-BASED MONITORING SYSTEM

An Arduino microcontroller is incorporated into the proposed system to monitor electrical parameters through suitable sensing circuits. The monitoring arrangement helps observe changes in generated voltage under different experimental operating conditions.

4. DC–DC VOLTAGE CONVERSION

A DC–DC converter can be used to condition the variable electrical output produced by the thermoelectric module. The converter helps provide a suitable regulated voltage when the available input voltage and power satisfy its operating requirements.

5.FUTURE AUTOMOTIVE APPLICATIONS

The proposed system provides a foundation for future automotive waste heat recovery applications. With high-temperature thermoelectric generator modules, improved heat exchangers, effective cooling systems, and suitable electrical protection.

CHAPTER 2
LITERATURE REVIEW

The following literature review is prepared for your project, “Design and Fabrication of a Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator.”

2.1 WASTE HEAT RECOVERY FROM INTERNAL COMBUSTION ENGINES

Burnete et al. (2022) conducted a comprehensive review of thermoelectric generation for waste heat recovery in internal combustion engines. The study identified engine exhaust gases and radiator cooling systems as major sources of rejected thermal energy. The researchers explained that thermoelectric generators can recover a portion of this wasted heat and convert it directly into electrical energy without requiring moving mechanical components. The study examined different heat exchanger arrangements, cooling methods, thermoelectric materials, and their effects on electrical output. It also highlighted the importance of reducing exhaust backpressure and maintaining suitable operating temperatures. The research concluded that thermoelectric generation is a promising technology for automotive energy recovery, although its practical performance depends on thermal management, material selection, and system integration.

2.2 THERMOELECTRIC GENERATOR MATERIALS AND PERFORMANCE

Bhakta and Kundu (2024) reviewed the development of thermoelectric generators for automobile waste heat recovery and improved energy utilization. The study discussed the importance of thermoelectric materials, semiconductor properties, module geometry, and heat transfer arrangements in determining energy conversion efficiency. The researchers examined different methods for improving electrical output, including advanced thermoelectric materials, improved heat exchanger configurations, and modifications to thermoelectric module structures. The study emphasized that the performance of a thermoelectric generator depends strongly on the temperature difference maintained between its hot and cold surfaces. It concluded that improvements in material properties, thermal management, and structural design are necessary to increase the effectiveness of automotive thermoelectric generators. This research provides a useful

foundation for selecting and evaluating thermoelectric modules in experimental waste heat recovery systems.

2.3 HEAT TRANSFER ENHANCEMENT IN THERMOELECTRIC GENERATORS

Patil et al. (2024) investigated the influence of exhaust heat exchanger inserts on the performance of thermoelectric generators. The researchers developed an experimental arrangement using an exhaust heat exchanger, thermoelectric modules, and a water-based cooling system. Different internal insert configurations were tested to improve the transfer of thermal energy from exhaust gases to the hot side of the thermoelectric modules. The experimental results showed that cone-shaped and cone-cylinder inserts improved electrical power generation compared with a heat exchanger without inserts. The cone-shaped configuration increased the average maximum electrical output by approximately 29.49%, while the cone-cylinder arrangement produced an improvement of approximately 26.11%. The study demonstrated that heat exchanger geometry and thermal contact significantly influence thermoelectric generator performance. These findings highlight the importance of effective heat collection and temperature difference management in waste heat recovery systems.

2.4 THERMAL MANAGEMENT AND COOLING ARRANGEMENTS

Sukarno et al. (2025) experimentally investigated a thermoelectric generator system assisted by heat pipe heat sinks for recovering waste heat from a pickup vehicle exhaust. The researchers installed thermoelectric generator units on a heat-collecting block connected to the exhaust pipe and evaluated different module arrangements and cooling conditions. The study demonstrated that heat pipe cooling combined with forced convection improved heat dissipation from the cold side and increased the temperature difference across the thermoelectric modules. Under the reported experimental conditions, the system achieved a maximum electrical output of approximately 1.4 W and a thermoelectric efficiency of 1.14% at an exhaust pipe temperature of 180°C. The researchers concluded that effective cooling is essential for improving electrical power generation. This study supports the use of an aluminium heat sink in the proposed prototype and indicates that more advanced cooling arrangements may improve future system performance.

2.5 AUTOMOTIVE THERMOELECTRIC GENERATOR OPTIMIZATION

Zhao et al. (2024) investigated the energy and exergy performance of a thermoelectric generator system designed for automotive exhaust waste heat recovery. The researchers developed a thermodynamic model to study the influence of exhaust temperature, heat transfer characteristics, and internal energy losses on thermoelectric conversion. The study identified significant energy losses during heat transfer from exhaust gases and through the thermoelectric elements. It also showed that improving the exhaust-side heat transfer coefficient can increase the overall exergy efficiency of the system. The researchers emphasized that higher exhaust temperatures do not automatically produce better overall thermodynamic performance because additional losses may occur. The study concluded that effective heat exchanger design, temperature management, and reduction of internal thermal losses are important for improving automotive thermoelectric generator performance. These findings provide guidance for developing and evaluating the proposed waste heat recovery prototype.

CHAPTER 3
PROBLEM IDENTIFICATION

3.1 LOSS OF THERMAL ENERGY THROUGH EXHAUST GASES

Internal combustion engines convert only a portion of the chemical energy supplied by fuel into useful mechanical energy. A considerable amount of energy is rejected as heat through exhaust gases, engine cooling systems, and other components. The exhaust gases leave the engine at elevated temperatures and carry valuable thermal energy directly into the atmosphere. In conventional vehicles, much of this exhaust heat is not recovered for useful applications. This results in reduced overall energy utilization and represents an opportunity for energy recovery. Therefore, an effective waste heat recovery system is required to convert a portion of the available exhaust thermal energy into useful electrical power.

3.2 LIMITED CONVERSION OF WASTE HEAT INTO USEFUL ENERGY

Conventional automotive electrical systems generally depend on engine-driven alternators or other electrical generation arrangements to supply power to auxiliary components. These systems consume a portion of the available mechanical energy to generate electricity. Although several waste heat recovery technologies have been developed, their application in small automotive systems can be limited by cost, complexity, installation space, and operating requirements. Thermoelectric generators provide an alternative method of converting thermal energy directly into electricity without moving mechanical components. However, their relatively low conversion efficiency and limited electrical output remain important challenges.

3.3 DIFFICULTY IN MAINTAINING A TEMPERATURE DIFFERENCE

The performance of a thermoelectric generator mainly depends on the temperature difference maintained between its hot and cold surfaces. During operation, heat is transferred from the exhaust heat source to the hot side of the thermoelectric module. At the same time, heat must be removed from the cold side to maintain the required temperature gradient. If the cold-side temperature increases, the temperature difference decreases, resulting in reduced electrical output. Insufficient cooling, poor thermal contact, and variations in heat source temperature can affect system performance.

3.4 LIMITATIONS OF THERMOELECTRIC MODULE PERFORMANCE

Thermoelectric modules have certain limitations that affect their suitability for automotive waste heat recovery applications. The electrical output generated by a thermoelectric module depends on its material properties, temperature difference, internal electrical resistance, and connected load. The TEC1-12706 module used in the proposed prototype is primarily designed for thermoelectric cooling applications and is not intended for direct exposure to high-temperature engine exhaust. Excessive heating may damage the semiconductor elements and soldered connections within the module. Therefore, suitable thermal protection and temperature control are essential during experimental testing.

3.5 CHALLENGES IN ELECTRICAL OUTPUT MEASUREMENT

The electrical voltage generated by a thermoelectric module varies according to the temperature difference and operating conditions. During low-temperature-gradient operation, the generated voltage may be relatively small and insufficient to directly power conventional electrical devices. Accurate measurement is therefore important for evaluating system performance. Electrical monitoring requires suitable voltage measurement circuits and instruments capable of recording small variations in output. An Arduino-based monitoring system can be incorporated to observe the generated voltage under different experimental conditions. However, voltage measurement alone is insufficient to determine usable electrical power.

3.6 THERMAL PROTECTION AND COOLING REQUIREMENTS

Effective thermal management is one of the major challenges in designing a thermoelectric waste heat recovery system. The hot side of the thermoelectric module must receive sufficient thermal energy, while the cold side must be maintained at a lower temperature. Improper heat transfer arrangements can result in excessive module temperature, reduced electrical output, or permanent damage to the thermoelectric device. In addition, direct exposure to engine exhaust temperatures may exceed the safe operating limits of a TEC1-12706 module.

CHAPTER 4
OBJECTIVE

4.1 GENERAL OBJECTIVE

The main objective of this project is to design and fabricate a **Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator** to demonstrate the conversion of wasted thermal energy into useful electrical energy. The proposed system aims to utilize the temperature difference between a heat source and a cooling arrangement to generate direct-current (DC) electrical voltage through the Seebeck effect.

The project focuses on developing a simple, compact, and low-cost experimental prototype using a TEC1-12706 thermoelectric module, and an Arduino-based monitoring system. The system is intended to investigate the possibility of recovering a portion of the thermal energy that would otherwise be discharged through the exhaust gases of internal combustion engines.

The project also aims to study the relationship between temperature difference and electrical output, evaluate the performance of the thermoelectric module under experimental conditions, and identify suitable methods for improving heat transfer and cooling efficiency. The overall objective is to demonstrate the potential of thermoelectric energy conversion as an auxiliary power generation technology and establish a foundation for future automotive waste heat recovery applications.

4.2 SPECIFIC OBJECTIVES

The specific objectives of the proposed project are as follows:

1. To design and fabricate a waste heat recovery prototype using a thermoelectric module for converting thermal energy into electrical energy.
2. To utilize the Seebeck effect for generating direct-current electrical voltage by maintaining a temperature difference between the hot and cold sides of the thermoelectric module.
3. To incorporate a TEC1-12706 thermoelectric module for experimentally demonstrating thermoelectric energy conversion under suitable operating temperatures.

4. To develop an effective heat collection arrangement for transferring thermal energy from the heat source to the hot side of the thermoelectric module.
5. To integrate an Arduino-based monitoring system for observing and recording the electrical output generated by the thermoelectric module using appropriate measurement circuits.
6. To investigate the use of a DC–DC converter for conditioning the generated electrical voltage according to the requirements of a suitable low-power electrical load.
7. To study the relationship between temperature difference and electrical output by conducting experimental tests under different heating and cooling conditions.
8. To evaluate the electrical performance of the prototype by measuring generated voltage and, where suitable measurement arrangements are available, current and output power.
9. To develop a compact, economical, and reliable experimental arrangement using commonly available mechanical and electronic components.
10. To identify the thermal and electrical limitations of the proposed system and determine the improvements required for practical engine exhaust heat recovery.
11. To explore the potential application of thermoelectric waste heat recovery for auxiliary electrical power generation in internal combustion engine and hybrid vehicles.

CHAPTER 5
SELECTION OF MATERIALS

The components plays an important role in the design and fabrication of a Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator. The components are selected based on their thermal properties, electrical characteristics, availability, cost, durability, and suitability for experimental applications.

5.1 THERMOELECTRIC MODULE (TEC1-12706)

The TEC1-12706 thermoelectric module is the primary energy conversion component used in the proposed waste heat recovery system. It is a semiconductor-based device that operates using thermoelectric principles. Although the TEC1-12706 is commonly designed for cooling applications based on the Peltier effect, it can also generate electrical voltage when a temperature difference is maintained between its two surfaces through the Seebeck effect.

The module consists of multiple pairs of P-type and N-type semiconductor elements connected electrically in series and thermally in parallel. These semiconductor elements are enclosed between two ceramic plates that provide electrical insulation and support heat transfer. When one surface is heated and the opposite surface is maintained at a lower temperature, charge carriers move through the semiconductor elements, producing a direct-current voltage across the electrical terminals.

The TEC1-12706 module is selected for the experimental prototype because of its compact size, simple construction, low cost, and easy availability. The commonly available module has approximate dimensions of 40 mm × 40 mm × 3.8 mm, although specifications may vary between manufacturers. Its performance depends on the temperature difference, thermal contact, and connected electrical load.

5.2 ARDUINO MICROCONTROLLER

The Arduino microcontroller is selected as the monitoring and control platform for the proposed waste heat recovery system. Arduino boards are widely used in engineering projects because of their simple programming environment, availability, compatibility with sensors, and ability to process electrical signals.

The Arduino can receive voltage signals from suitable measurement circuits and process the collected information using a programmed sequence. The measured parameters can be displayed on an LCD or transmitted to a computer through serial communication for observation and recording.

In the proposed system, Arduino-based monitoring is intended to assist in evaluating the electrical output generated by the thermoelectric module. The monitoring arrangement can be expanded to include temperature measurement, voltage monitoring, and other experimental parameters depending on the available sensors.

The Arduino is selected because it provides a flexible and economical platform for developing an experimental monitoring system. However, the thermoelectric module may not generate sufficient power to operate the Arduino directly. Therefore, a separate regulated power supply may be required during laboratory testing.

5.3 TEMPERATURE SENSORS

Temperature sensors are important measurement components in a thermoelectric waste heat recovery system. The electrical output generated by a thermoelectric module depends on the temperature difference between its hot and cold surfaces. Therefore, accurate temperature measurement is necessary for evaluating the thermal operating conditions of the prototype.

Suitable temperature sensors can be positioned near the hot-side and cold-side surfaces of the thermoelectric module. These sensors provide temperature readings that can be used to calculate the temperature difference across the module.

Temperature monitoring helps identify changes in operating conditions and supports the safe evaluation of the thermoelectric module.

5.4 VOLTAGE MEASUREMENT CIRCUIT

The voltage measurement circuit is used to measure the direct-current electrical voltage generated by the thermoelectric module. The output voltage varies according to the temperature difference, module characteristics, and connected electrical load.

A suitable voltage measurement arrangement can be connected across the output terminals of the thermoelectric module. The generated voltage may be measured using a digital multimeter or an Arduino-compatible voltage sensing circuit.

When an Arduino is used, the voltage signal must remain within the permitted input range of the microcontroller. A voltage divider or suitable signal-conditioning circuit may be required depending on the expected output voltage.

Accurate voltage measurement is important for evaluating the performance of the thermoelectric module under different experimental conditions. The voltage readings can be compared with the measured temperature difference to study the relationship between thermal input conditions and electrical generation.

For a complete power evaluation, the voltage across a known electrical load and the corresponding current must also be measured.

5.5 LCD DISPLAY

A Liquid Crystal Display (LCD) can be incorporated into the monitoring system to display measured electrical and thermal parameters. LCD modules are commonly used in electronic projects because of their low power consumption, simple interface, compact size, and clear numerical display.

A typical character LCD can display information such as generated voltage, hot-side temperature, cold-side temperature, and temperature difference when the required sensors and programming are provided.

The display is connected to the Arduino through suitable electrical connections. The microcontroller processes the measured signals and updates the displayed values according to the programmed instructions.

An LCD display provides convenient observation of experimental parameters without requiring continuous connection to a computer. It can therefore improve the usability of the monitoring system during laboratory demonstrations.

5.6 DC–DC CONVERTER

A DC–DC converter is an electrical power-conditioning component used to convert one direct-current voltage level into another. The output voltage generated by a thermoelectric module varies with the available temperature difference and connected load. Therefore, a suitable converter may be required when the generated electrical energy is intended for a specific application.

For thermoelectric applications, the converter must be selected according to the minimum input voltage, available electrical power, conversion efficiency, and load requirements. Conventional boost converters may not start or operate efficiently when supplied with the very low voltage generated by a single thermoelectric module.

The DC–DC converter is considered for the proposed system to investigate the possibility of obtaining a regulated electrical output. Its practical operation must be verified using actual experimental voltage and current measurements.

5.7 ELECTRICAL CONNECTIONS AND WIRING

Electrical connections and wiring are necessary for connecting the thermoelectric module, monitoring circuit, Arduino, display, and power-conditioning components. Proper wiring ensures reliable electrical signal transmission and helps reduce unwanted electrical resistance and connection losses.

Insulated copper wires are commonly used because of their good electrical conductivity, flexibility, and availability. The wire size and insulation must be selected according to the expected current, voltage, and operating temperature.

Connections near the heat source require additional care because ordinary wire insulation may be damaged by excessive heat. Suitable high-temperature wiring or protective routing should be used where necessary.

Secure terminals, connectors, and properly insulated joints help prevent loose connections and accidental short circuits. Correct polarity must also be maintained when connecting the thermoelectric module to the measurement and power-conditioning circuits.

5.8 SUPPORTING FRAME AND ASSEMBLY

The supporting frame provides mechanical stability to the experimental waste heat recovery system. It holds the thermoelectric module, heat collection arrangement, aluminium heat sink, and associated components in their required positions.

The supporting structure should be strong enough to maintain component alignment and withstand the expected experimental conditions. Suitable metallic brackets, plates, and fasteners may be used for constructing the frame.

The design must provide sufficient clearance around heated components and allow convenient access for measurement and maintenance. The frame should also prevent direct contact between heat-sensitive electrical components and the hot-side arrangement.

A compact and stable supporting structure improves the safety, handling, and repeatability of experimental testing. Proper mechanical assembly is therefore important for the successful demonstration of the proposed thermoelectric waste heat recovery system.

5.9 POWER SUPPLY AND PROTECTION COMPONENTS

A suitable power supply is required to operate the Arduino microcontroller and associated monitoring components during experimental testing. Since the electrical output of a single TEC1-12706 module may be insufficient to power the monitoring system, an external regulated DC supply or USB power source can be used.

Electrical protection components may include suitable fuses, switches, insulated connectors, and voltage protection circuits. These components help reduce the risk of electrical damage caused by incorrect connections, excessive current, or unexpected voltage variations.

Thermal protection is equally important because excessive temperature can damage the thermoelectric module and nearby electrical components. Temperature monitoring, proper heat source positioning, and adequate cooling should therefore be considered during prototype operation.

CHAPTER 6

COMPONENTS

6.1 COMPONENTS

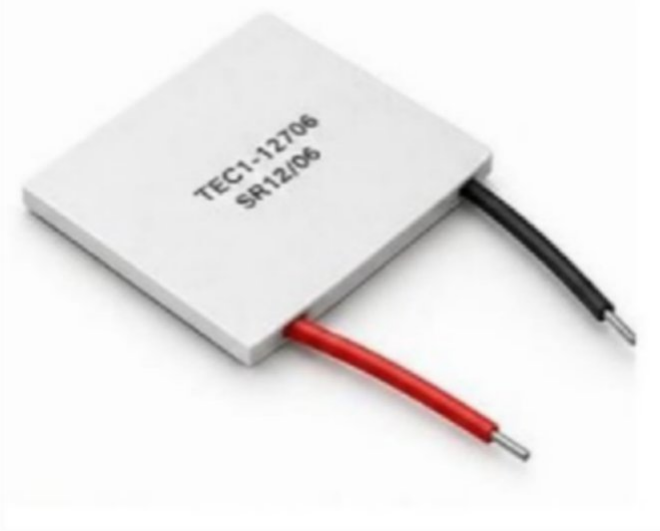


Fig 6.1 Thermoelectric generator (teg)



Fig 6.2 Arduino uno board



Fig 6.3 DC – DC Converter

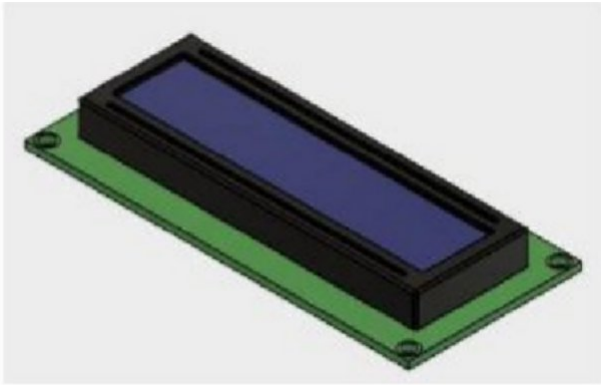


Fig 6.4 LCD Display

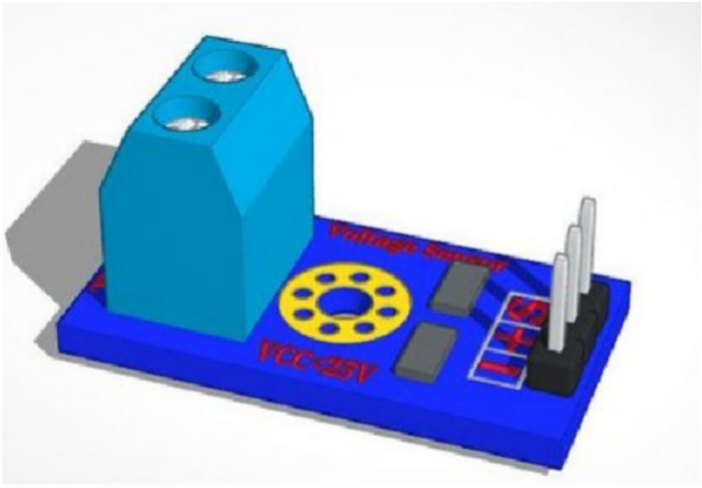


Fig 6.5 Voltage sensor

CHAPTER 7

METHODOLOGY

7.1 WORKING METHOD

The proposed waste heat recovery system works on the principle of the Seebeck effect, which converts a temperature difference directly into electrical voltage.

Heat Collection

Thermal energy from the heat source is transferred to the heat collection plate. In the preliminary experimental setup, a candle flame is used to demonstrate the heating process. The collected heat is conducted to the hot side of the thermoelectric module.

Formation of Temperature Difference

The aluminium heat sink is attached to the cold side of the thermoelectric module. It absorbs heat and releases it to the surrounding atmosphere. As a result, a temperature difference is maintained between the hot and cold surfaces of the module.

Electrical Voltage Generation

Due to the Seebeck effect, the temperature difference across the thermoelectric module produces a direct-current electrical voltage. The generated voltage depends mainly on the temperature difference, module characteristics, and thermal contact conditions.

Voltage Measurement and Monitoring

The generated voltage is measured using a digital multimeter or a suitable voltage measurement circuit. Temperature sensors and an Arduino-based monitoring system can be used to observe the operating conditions and electrical output.

Experimental Observation

During preliminary laboratory testing, an output voltage of approximately **0.2 V to 0.3 V** was observed. This demonstrates the basic principle of converting thermal energy into electrical voltage using a thermoelectric module.

Electrical Power Utilization

The generated voltage may be supplied to a suitable DC–DC converter for voltage regulation if sufficient input voltage and power are available. With further improvements, the recovered electrical energy could potentially be used for compatible low-power applications.

7.2 WORKING METHOD FLOWCHART

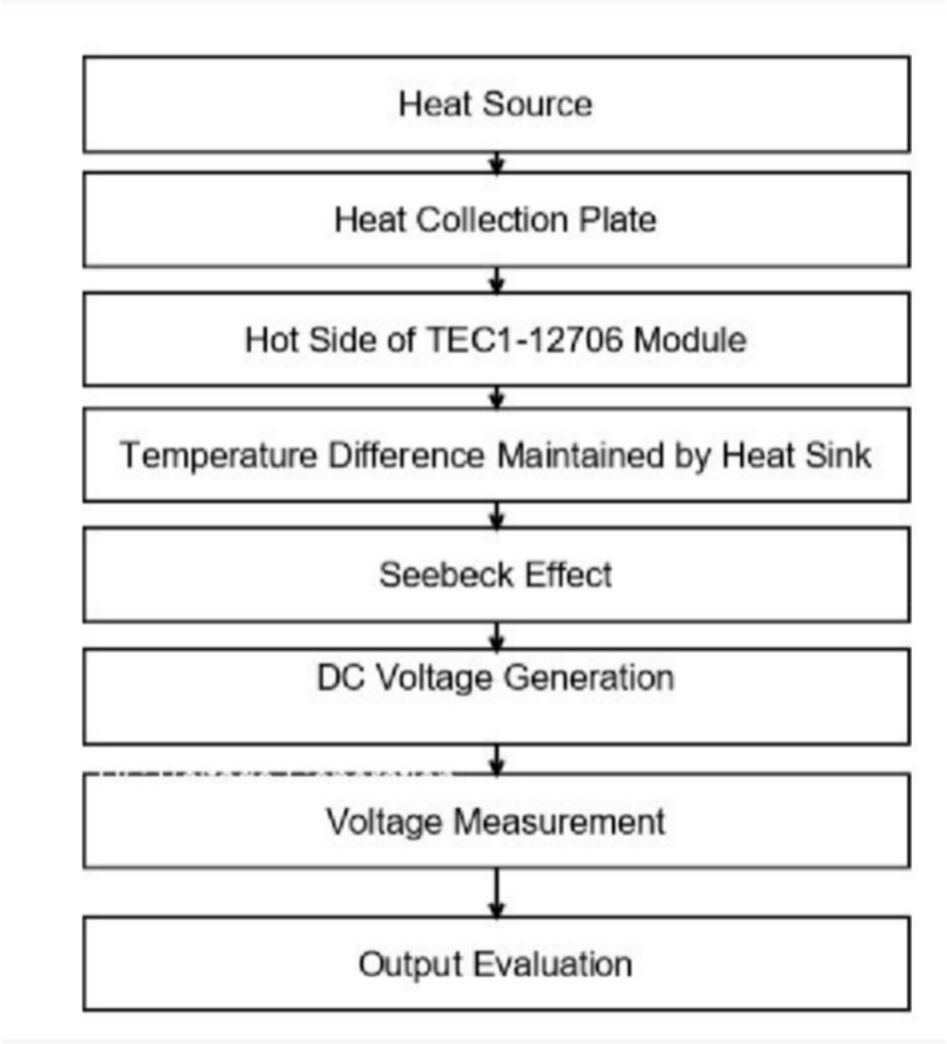


Fig 7.1 Working method flowchart

CHAPTER 8
COST ESTIMATION

8.1 BILL OF MATERIALS

The cost estimation for the proposed Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator is prepared based on the components required for the experimental prototype. The major components include the TEC1-12706 thermoelectric module, Arduino microcontroller, temperature sensors, DC–DC converter, LCD display, and supporting materials.

8.2 TOTAL COST ESTIMATION

Table 8.1 Total cost estimation

Component / Material	Purpose	Quantity	Unit Cost (₹)	Total Cost (₹)
TEC1-12706 Thermoelectric Module	Converts heat into electrical energy	1 Units	200	200
Arduino UNO	Monitoring system	1 Unit	450	450
Temperature Sensors	Temperature measurement	1 Units	150	150
LCD Display (16×2)	Displays measured values	1 Unit	180	180
DC–DC Boost Converter	Voltage conditioning	1 Unit	150	150
Voltage Sensor Module	Measures generated voltage	1 Unit	100	100
Connecting Wires	Electrical connections	1 Set	150	150
TOTAL COST ESTIMATION				**2350**

CHAPTER 9

WORKING PROCESS

9.1 OVERVIEW OF THE WORKING PROCESS

The Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator operates on the principle of converting thermal energy into electrical energy through the Seebeck effect. The proposed system is designed to utilize waste heat from engine exhaust gases and convert a portion of this thermal energy into useful direct-current (DC) electrical power.

During operation, thermal energy from the heat source is transferred to the hot side of the thermoelectric module through a heat-collecting surface. The cold side is connected to an aluminium heat sink, which dissipates heat into the surrounding atmosphere. This arrangement creates a temperature difference between the two surfaces of the thermoelectric module. Due to the Seebeck effect, the temperature difference produces an electrical voltage across the module terminals.

The generated electrical output is measured using suitable instruments and can be monitored through an Arduino-based system. The voltage varies depending on the temperature difference, heat transfer conditions, and connected electrical load. A suitable DC–DC converter may be incorporated to regulate the generated voltage for low-power applications. The proposed prototype demonstrates the basic working process of thermoelectric energy conversion and provides a foundation for future automotive waste heat recovery applications.

9.2 STEP-BY-STEP OPERATION

The working process of the proposed waste heat recovery system involves the following stages.

Step 1: Preparation of the Experimental Setup

Initially, the thermoelectric module is positioned between the heat-collecting surface and the aluminium heat sink. The components are properly aligned and secured using suitable mounting arrangements. Electrical connections are made between the thermoelectric module and the voltage measurement circuit. The monitoring components are checked before starting the experiment.

Step 2: Application of Thermal Energy

A controlled heat source is used to supply thermal energy to the hot-side heat collection arrangement. In the preliminary experimental model, a candle flame is used to demonstrate the heat transfer process. The heat is transferred through the metallic surface to the hot side of the thermoelectric module. Direct flame contact with the module is avoided to prevent thermal damage.

Step 3: Cooling of the Cold Side

This cooling process helps maintain a temperature difference between the hot and cold surfaces. The effectiveness of the heat sink depends on its surface area, thermal conductivity, and surrounding airflow conditions.

Step 4: Generation of Electrical Voltage

When a temperature difference is established across the thermoelectric module, the Seebeck effect produces a direct-current voltage. The semiconductor elements inside the module convert part of the transferred thermal energy into electrical energy. The generated voltage increases or decreases according to the operating temperature difference and module characteristics.

Step 5: Electrical Output Measurement

The electrical output generated by the thermoelectric module is measured using a digital multimeter or a suitable voltage measurement circuit. The voltage readings are recorded under different heating conditions. Current measurement can also be performed using an appropriate electrical load to determine the actual electrical power generated by the module.

Step 6: Arduino-Based Monitoring

The generated electrical signals can be supplied to the Arduino microcontroller through suitable sensing circuits. The Arduino processes the measured values and can display them on an LCD or transmit them to a computer through serial communication. This arrangement helps observe changes in electrical output during experimental testing.

Step 7: Voltage Conditioning

The generated voltage may be supplied to a suitable DC–DC converter for electrical power conditioning. The converter is used to obtain a regulated voltage when the thermoelectric module provides sufficient input voltage and power. The conditioned output may then be used for compatible low-power electrical applications.

Step 8: Recording of Experimental Observations

The measured voltage, current, and temperature values are recorded during the experiment. These observations are used to evaluate the relationship between temperature difference and electrical output. The recorded data also help identify the limitations of the prototype and possible methods for improving performance.

9.3 ELECTRICAL OUTPUT MEASUREMENT

The electrical output measurement is an important part of evaluating the performance of the proposed thermoelectric waste heat recovery system. The electrical voltage generated by the TEC1-12706 module depends on the temperature difference between its hot and cold surfaces.

During experimental operation, the voltage is measured across the output terminals of the thermoelectric module. The measured values can be recorded at different temperature differences to observe the effect of heating and cooling conditions on electrical generation.

The electrical power generated by the thermoelectric module is calculated using the following equation:

$$\mathbf{P = V \times I}$$

Where,

P = Electrical power generated (W)

V = Output voltage across the load (V)

I = Electrical current through the load (A)

The temperature difference is calculated using:

$$\Delta T = Th - Tc$$

Where,

ΔT = Temperature difference (°C)

Th = Hot-side temperature (°C)

Tc = Cold-side temperature (°C)

The electrical output can be evaluated by comparing the generated voltage and power at different temperature differences. Accurate measurement of voltage and current under load is necessary to determine the usable electrical power. Open-circuit voltage alone cannot represent the actual power delivered by the system.

9.4 APPLICATIONS AND LIMITATIONS

The proposed thermoelectric waste heat recovery system demonstrates the potential of converting wasted thermal energy into useful electrical energy. With suitable modifications, the technology may be applied in automotive exhaust systems, industrial heating equipment, furnaces, boilers, and other applications where a temperature difference is available.

In automotive applications, thermoelectric generators may be used to recover a portion of exhaust thermal energy and supply electrical power to compatible auxiliary systems. The absence of moving parts makes thermoelectric generation attractive for applications requiring compact construction and low mechanical maintenance.

However, the proposed experimental prototype has certain limitations. The TEC1-12706 module is primarily designed for thermoelectric cooling and is not suitable for direct exposure to high-temperature engine exhaust. The electrical output of a single module is relatively small and depends on the temperature difference and connected load. Maintaining effective cooling and selecting a suitable DC–DC converter are additional challenges.

9.5 CIRCUIT DIAGRAM

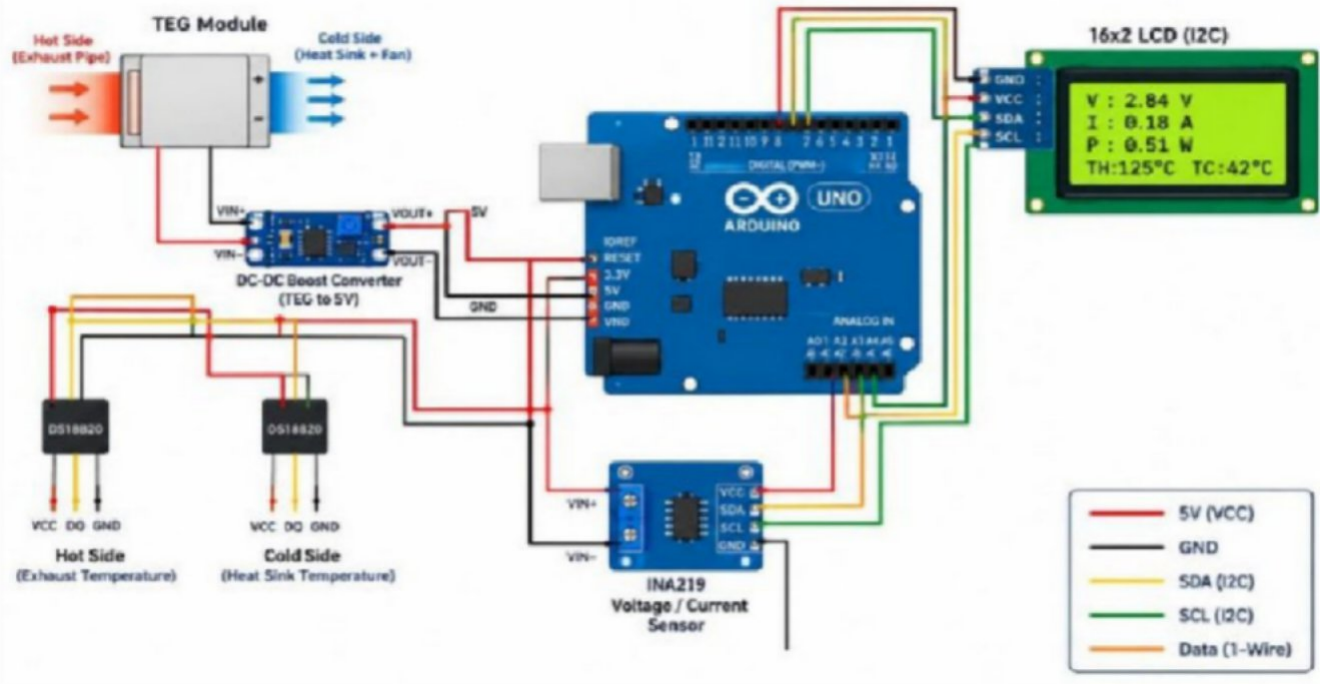


Fig 9.1 Circuit diagram

9.6 STEP BY STEP PROCESS

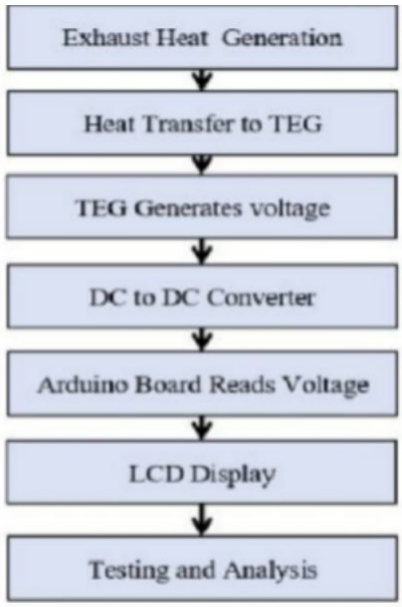


Fig 9.2 Step by step process

CHAPTER 10
FABRICATED MODEL

10.1 FABARICATED MODEL

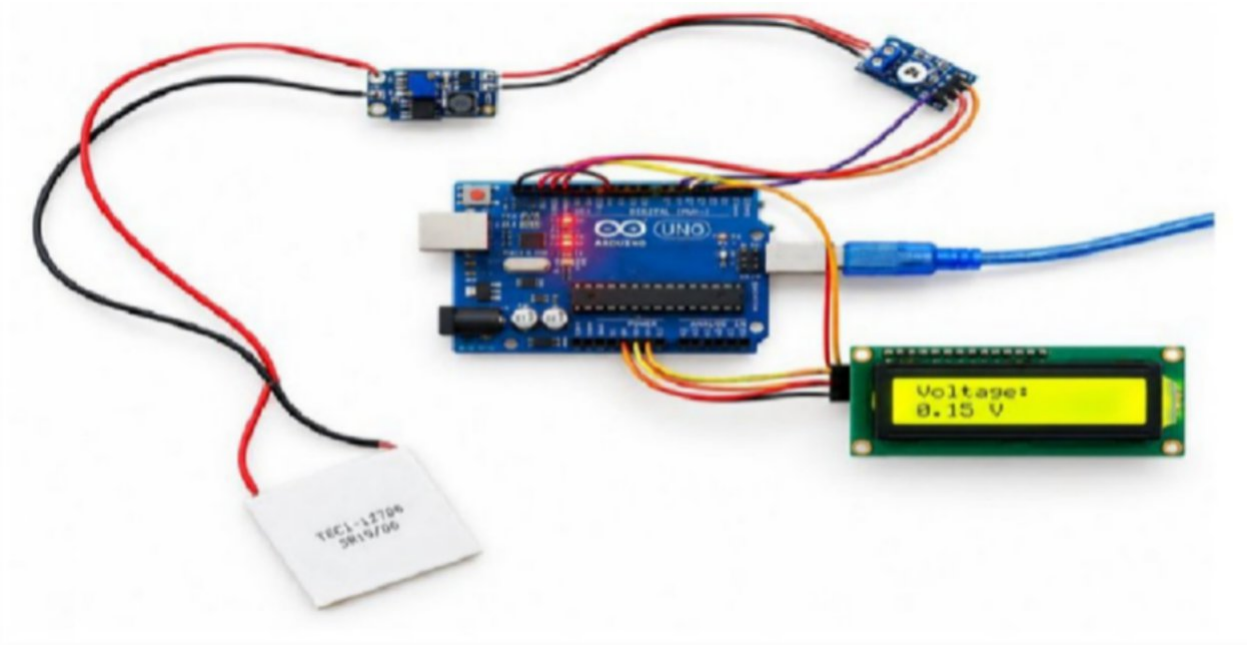


Fig 10.1 Fabricated model

CHAPTER 11

DESIGN CALCULATION

11.1 TEMPERATURE DIFFERENCE CALCULATION

Formula:

$$\Delta T = Th - Tc$$

Given:

Hot-side temperature (Th) = 60°C

Cold-side temperature (Tc) = 35°C

Calculation:

$\Delta T = 60 - 35$

$\Delta T = 25°C$

Result: Temperature Difference = 25°C

11.2 SEEBECK VOLTAGE CALCULATION

Formula:

$$Voc = \alpha \times \Delta T$$

Given:

Seebeck coefficient (α) = 0.05 V/K

Temperature difference (ΔT) = 25 K

Calculation:

$Voc = 0.05 \times 25$

$Voc = 1.25$ V

Result: Seebeck Voltage = 1.25 V

11.3 ELECTRICAL POWER CALCULATION

Formula:

$$P = V \times I$$

Given:

Output Voltage (V) = 0.25 V

Output Current (I) = 0.10 A

Calculation:

$P = 0.25 \times 0.10$

$P = 0.025$ W

$P = 25$ mW

Result: Electrical Power = 25 mW

11.4 MAXIMUM POWER OUTPUT CALCULATION

Formula:

$$P_{max} = V_{oc}^2 / (4 \times R_{int})$$

Given:

Open-circuit voltage (Voc) = 1.25 V

Internal resistance (Rint) = 2 Ω

Calculation:

$P_{max} = (1.25 \times 1.25) / (4 \times 2)$

$P_{max} = 1.5625 / 8$

$P_{max} = 0.1953$ W

Result: Maximum Power Output = 0.1953 W

11.5 THERMOELECTRIC EFFICIENCY CALCULATION

Formula:

$$\eta = (P_{out} / Q_{in}) \times 100$$

Given:

Electrical Output Power (Pout) = 0.025 W

Thermal Input Power (Qin) = 5 W

Calculation:

$\eta = (0.025 / 5) \times 100$

$\eta = 0.005 \times 100$

$\eta = 0.5\%$

Result: Thermoelectric Efficiency = 0.5%

11.6 CALCULATION RESULT

Table 11.1 Calculation result

S. No.	Parameter	Calculated Value
1	Temperature Difference	25°C
2	Seebeck Voltage	1.25 V
3	Electrical Output Power	25 mW
4	Maximum Theoretical Power	0.1953 W
5	Thermoelectric Efficiency	0.5%

CHAPTER 12

CONCLUSION

12.1 CONCLUSION OF THE PROJECT

The project titled “Design and Fabrication of a Waste Heat Recovery System from Engine Exhaust Using Thermoelectric Generator” was undertaken to study the possibility of converting waste thermal energy into useful electrical energy through thermoelectric technology. The main objective of this project was to demonstrate the principle of waste heat recovery using a thermoelectric module and to understand the factors influencing electrical power generation.

The proposed system is based on the Seebeck effect, in which an electrical voltage is produced when a temperature difference is maintained between the hot and cold surfaces of a thermoelectric module.This temperature difference enables the generation of electrical voltage without any moving mechanical parts.

During the preliminary experimental demonstration, a candle flame was used as the heat source to simulate the basic heat transfer process. An output voltage of approximately 0.2 V to 0.3 V was observed, demonstrating the ability of the thermoelectric module to generate electrical voltage from a temperature difference.

The design calculations provided a theoretical understanding of temperature difference, Seebeck voltage, electrical power, maximum power transfer, and thermoelectric conversion efficiency. These calculations helped explain the relationship between thermal conditions and electrical generation. However, the calculated values were based on assumed parameters, and further experimental measurements of current, temperature, and heat input are required to determine the actual power output and efficiency of the prototype.

The project demonstrates several advantages of thermoelectric energy conversion, including compact construction, silent operation, simple maintenance, and the absence of moving parts. At the same time, it identifies important limitations such as low electrical output, dependence on temperature difference, and the need for efficient cooling. The TEC1-12706 module used in the preliminary prototype is not designed for direct exposure to high-temperature engine exhaust, so

a suitable high-temperature thermoelectric generator would be required for practical automotive applications.

In conclusion, the project successfully demonstrates the basic principle of converting waste heat into electrical voltage using thermoelectric technology at the laboratory level. Although the present prototype has not been tested on an actual engine exhaust system, it provides a useful foundation for understanding waste heat recovery and thermoelectric power generation. With improvements in thermoelectric materials, heat exchanger design, cooling arrangements, and electrical power management, this concept has potential for further development toward energy recovery applications in automobiles and industrial systems.

CHAPTER 13

FUTURE SCOPE

13.1 FUTURE SCOPE OF THE PROJECT

The waste heat recovery system using a thermoelectric generator has considerable scope for future development. The present prototype demonstrates the basic principle of converting a temperature difference into electrical energy using the Seebeck effect. Further improvements can enhance its electrical output, thermal performance, reliability, and suitability for practical applications.

1. **Application to Actual Engine Exhaust:** The system can be developed and tested using heat from an actual internal combustion engine exhaust. Suitable high-temperature thermoelectric generator modules and heat-resistant materials must be selected for safe operation.
2. **Improvement in Electrical Output:** Multiple thermoelectric modules can be connected in series or parallel to increase the output voltage or current, depending on the electrical requirements.
3. **Advanced Cooling System:** The cooling arrangement can be improved using larger heat sinks, forced-air cooling, or liquid cooling to maintain a greater temperature difference across the thermoelectric modules.
4. **Use of High-Performance Materials:** Advanced thermoelectric materials with better conversion efficiency and temperature resistance can replace the basic TEC1-12706 module for future engine-exhaust applications.
5. **Automatic Monitoring System:** An Arduino or other microcontroller can be used with temperature sensors and voltage and current sensors to continuously monitor temperature difference, voltage, current, and electrical power.
6. **Power Conditioning and Energy Storage:** An efficient DC–DC converter and a suitable energy-storage device, such as a rechargeable battery or supercapacitor, can be incorporated to store and regulate the generated electrical energy.
7. **Optimization of Heat Collection:** The design of the heat collection plate and heat exchanger can be improved to distribute heat uniformly across the thermoelectric modules and reduce thermal losses.
8. **Performance and Efficiency Analysis:** Future experiments can measure voltage, current, output power, heat input, and temperature difference under different operating conditions to

calculate the actual conversion efficiency.

9. **Automotive Applications:** After suitable design improvements and testing, the system may be investigated for supplying small auxiliary electrical loads in vehicles, thereby reducing some of the demand on the vehicle's electrical system.
10. **Environmental and Economic Benefits:** Further studies can evaluate the cost, durability, energy savings, and potential reduction in fuel consumption and emissions under realistic operating conditions.

In conclusion, the future development of this project should focus on improving the temperature difference, selecting suitable high-temperature modules, optimizing the heat exchanger and cooling system, and conducting practical engine-exhaust tests. The present prototype is a preliminary demonstration, and its suitability for automotive applications must be established through further testing and performance evaluation.

CHAPTER - 14

REFERENCES

14.1 BOOKS

1. Burnete, N.V., Mariasiu, F., Depcik, C., Barabas, I., & Moldovanu, D., Review of Thermoelectric Generation for Internal Combustion Engine Waste Heat Recovery, Progress in Energy and Combustion Science, Vol. 91, Article 101009, Elsevier, 2022.
2. Bhakta, S., & Kundu, B., A Review of Thermoelectric Generators in Automobile Waste Heat Recovery Systems for Improving Energy Utilization, Energies, Vol. 17, No. 5, Article 1016, MDPI, 2024.
3. Tien, T.N., Vu, Q.K., & Duy, V.N., Novel Designs of Thermoelectric Generator for Automotive Waste Heat Recovery: A Review, AIMS Energy, Vol. 10, No. 4, pp. 922–942, 2022.
4. Singh, R., Dogra, S., Dixit, S., Vatin, N.I., Bhardwaj, R., Sundramoorthy, A.K., et al., Advancements in Thermoelectric Materials for Efficient Waste Heat Recovery and Renewable Energy Generation, Hybrid Advances, Article 100176, Elsevier, 2024.

14.2 RESEARCH PAPERS / JOURNALS

5. Luo, D., Sun, Z., & Wang, R., Performance Investigation of a Thermoelectric Generator System Applied in Automobile Exhaust Waste Heat Recovery, Energy, Vol. 238, Article 121816, Elsevier, 2022.
6. Wang, R., Fu, Y., Luo, D., Chen, J., & Zhou, W., Performance Evaluation of an Automotive Thermoelectric Generator with a Non-Isometric Distributed Fin Heat Exchanger, Sustainable Energy & Fuels, Vol. 6, pp. 4446–4457, Royal Society of Chemistry, 2022.
7. Pujol, T., T'Jollyn, I., Massaguer, E., Massaguer, A., Cózar, I.R., & De Paepe, M., Design Optimization of Plate-Fin Heat Sink with Forced Convection for Single-Module Thermoelectric Generator, Applied Thermal Engineering, Vol. 221, Article 119866, Elsevier, 2023.

8. Chen, W.H., Uribe, M.C., Luo, D., Jin, L., Saw, L.H., & Lamba, R., Taguchi Optimization and Analysis of Variance for Thermoelectric Generators with Forced Convection Air Cooling, Applied Thermal Engineering, Vol. 231, Article 120878, Elsevier, 2023.
9. Akal, D., & Umut, İ., Monitoring of Electricity Generation from Exhaust Waste Heat and Wireless Data Recording from a Mobile Phone in Real Driving Conditions of a Vehicle, Environmental Science and Pollution Research, Vol. 30, No. 45, pp. 100576–100583, Springer, 2023.
10. Chen, C.Y., Du, K.W., Chung, Y.C., & Wu, C.I., Advancements in Thermoelectric Generator Design: Exploring Heat Exchanger Efficiency and Material Properties, Energies, Vol. 17, No. 2, Article 453, MDPI, 2024.
11. Babu, N.J., & Kumar, B.R., Optimization of Heat Sink Geometry for Improved Thermoelectric Generator Efficiency, International Journal of Heat and Technology, Vol. 42, No. 1, pp. 270–276, 2024.
12. Ionescu, V., Thermal, Energy and Exergy Analysis of Thermoelectric Generator System for Waste Heat Recovery Applications, Journal of Ecological Engineering, Vol. 25, No. 7, pp. 201–211, 2024.
13. Luo, D., Zhang, H., Cao, J., Yan, Y., & Cao, B., Innovative Design of an Annular Thermoelectric Generator for Enhanced Automotive Waste Heat Recovery, Energy Conversion and Management, Vol. 313, Article 118584, Elsevier, 2024.